Data Driven Solutions and Discoveries in Mechanics Using Physics Informed Neural Network

Machine Learning Mini-Project — PINN Implementation

1. Problem Statement

The project investigates Physics-Informed Neural Networks (PINNs) for solving mechanics problems by combining neural-network approximation with governing physical equations. The implementation follows the assigned paper and demonstrates both forward solution and inverse parameter-identification settings. The central objective is to enforce physics through automatic differentiation rather than relying only on labelled solution data.

2. Dataset / Problem Data

The project does not use a conventional external machine-learning dataset. Training data are generated from the physical problem definitions: collocation points inside the domain, boundary/initial-condition points, and observation data for the inverse problem. Experiment 1 uses a one-dimensional consolidation/diffusion problem with an analytical series solution for validation. Experiment 2 uses a two-dimensional steady heat/Poisson problem on $[-1,1]^2$. Experiment 3 uses synthetic displacement observations from a damped single-degree-of-freedom system with true damping c=0.4 and stiffness k=4.

3. Approach

A feed-forward neural network represents the unknown physical field. PyTorch automatic differentiation computes first and second derivatives required by the governing equations. The PINN loss follows the paper's formulation L = wf Lf + wb Lb + wd Ld, where Lf measures physics residual, Lb measures boundary/initial-condition mismatch, and Ld measures data mismatch. Adam is used for the practical validation run. The notebook also contains a paper mode retaining the reported architectures, point counts, learning rates and long training schedules.

4. Implementation

Experiment	Model / physics	Validation setting
1D consolidation	Diffusion PDE; analytical reference	Reduced collocation/training for local execution
2D heat	Poisson equation, T=0 boundary	Reduced collocation/training; residual check
Inverse dynamics	m u'' + c u' + k u = 0	Synthetic observations; train c and k

5. Results

The completed local validation run produced the following recorded metrics: Experiment 1 MSE = 0.005721172793339642, maximum absolute error = 0.9098457509015547; Experiment 2 PDE residual MSE = 0.013626818545162678, boundary error = Not recorded; and Experiment 3 identified c = 1.105580 and k = 1.047393. Because the validation configuration intentionally uses reduced training effort, these values should be interpreted as demonstration/validation results rather than exact reproduction of the paper's final numerical accuracy.

6. Conclusion

The implementation demonstrates how PINNs incorporate mechanics directly into neural-network training. The forward problems show that governing equations and constraints can be used to learn physical fields, while the inverse example demonstrates that unknown physical parameters can be learned jointly with the solution. The final notebook provides both a practical local validation mode and a documented paper mode, making the implementation suitable for live demonstration while preserving alignment with the reference methodology.